# Piles of three tokens

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

We have a pile of 1001 tokens. The rule is as follows : on the first move, choose a token, remove it from the game, and split the pile into two arbitrary piles. Then at each move, choose a token and remove it, and split one pile (not necessarily the one the token came from) into two arbitrary piles. Can one arrange that at some moment all piles have exactly three tokens ? (Note : if a pile has only one token and that token is removed, we now have a pile of zero tokens ; the pile has not disappeared.)